Measuring the Cost of Living

Primary Reference: Principles of Macroeconomics by Mankiw

Indraprastha Institute of Information Technology, Delhi

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- Prices generally rise over a period of time.
- This concept is know as _____?
- Some common examples?

Consumer Price Index

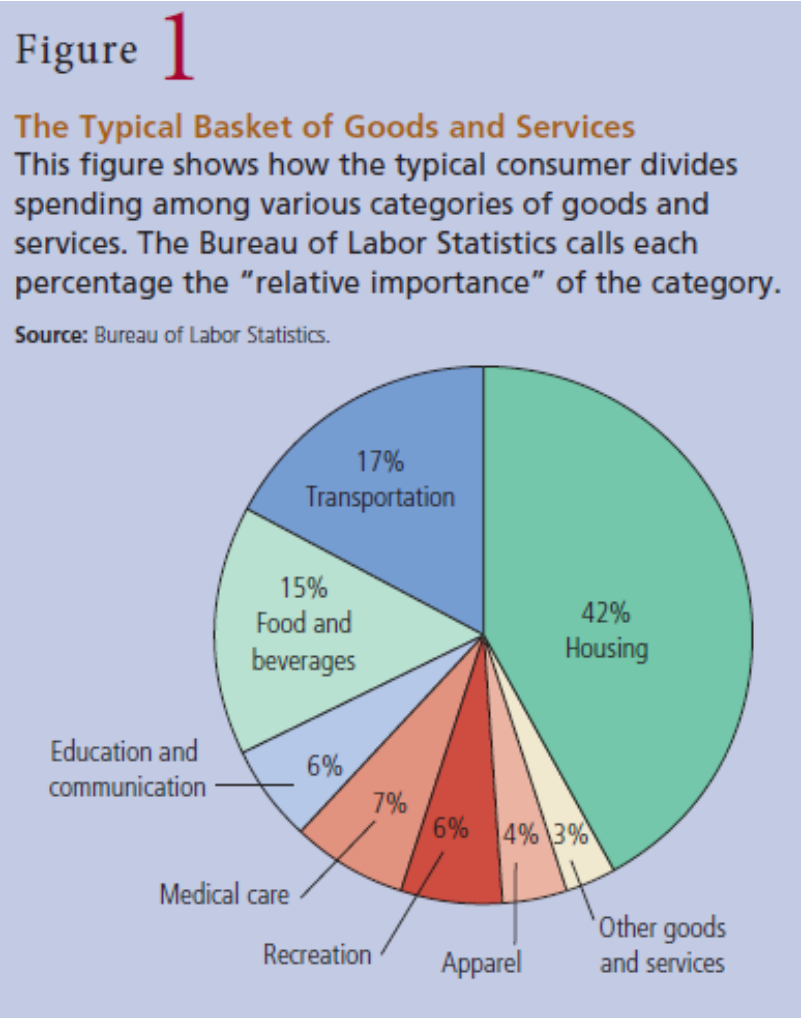
- Fix the basket.
- Find the prices.
- Compute the basket's cost.
- Choose a base year and compute the index.
- Compute the inflation rate.

Consumer Price Index

$$\text{Consumer Price Index} = \frac{\text{Price of basket of goods and services in current year}}{\text{Price of basket in base year}} \times 100$$

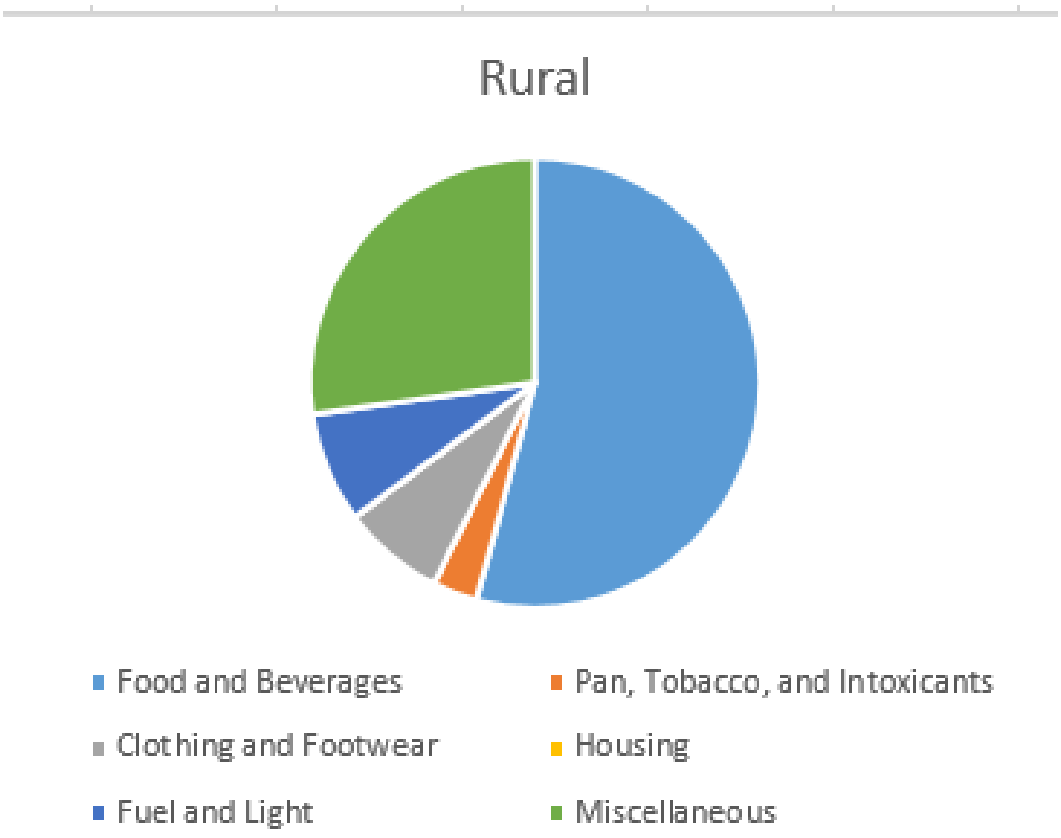
$$\text{Inflation rate in year 2} = \frac{\text{CPI in year 2} - \text{CPI in year 1}}{\text{CPI in year 1}}$$

What is in the CPI's Basket? (USA)



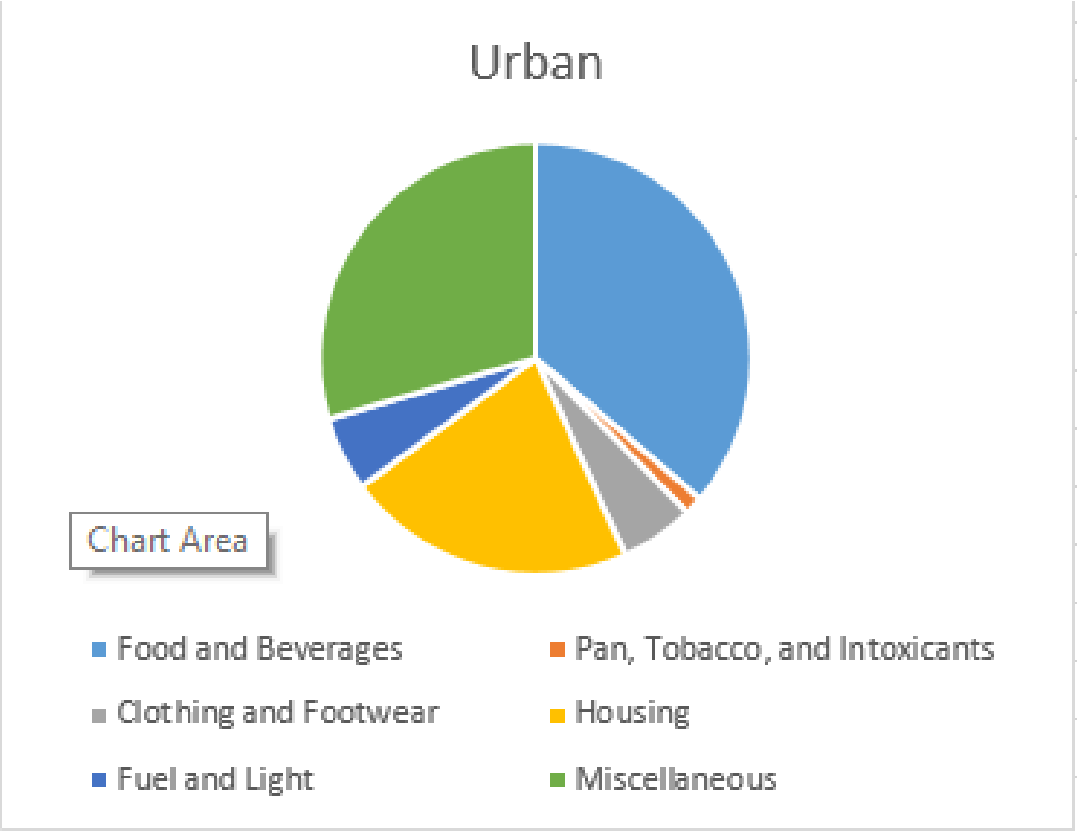
**Source: Principles of Macroeconomics by N. Gregory Mankiw.

What is in the CPI's Basket? (India)



**Source: Ministry of Statistics and Program Implementation (MOSPI). Click here for details.

What is in the CPI's Basket? (India)



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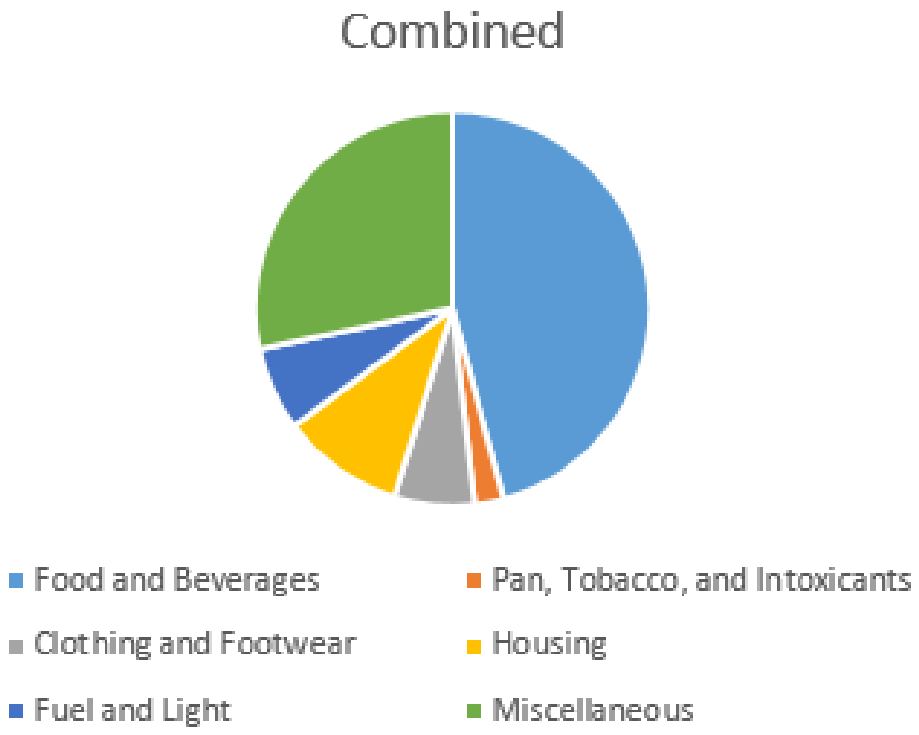
What is in the CPI's Basket? (India)



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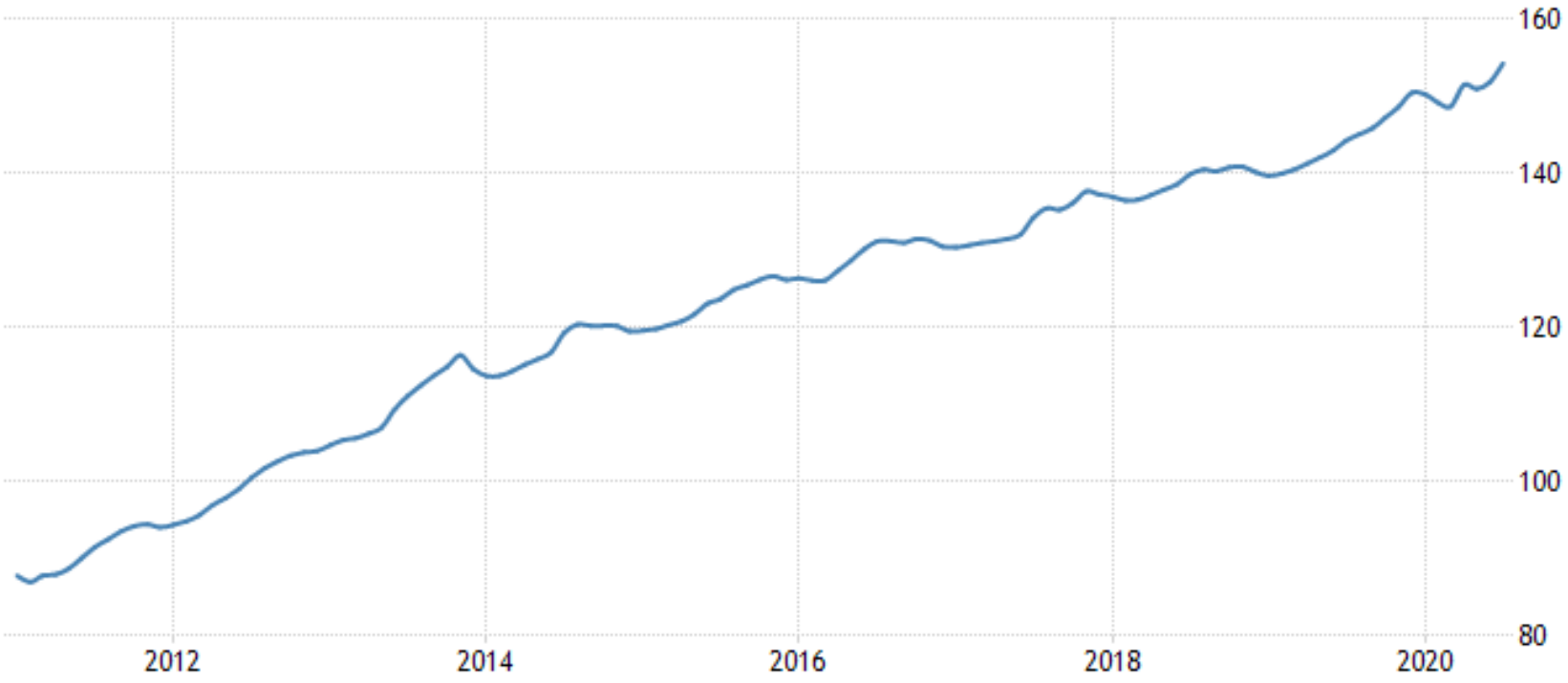
CPI over the years



**Source: Tradingeconomics.com | Ministry of Statistics and Program Implementation (MOSPI).
Click here for details.

Problems in Measuring the Cost of Living

- Substitution bias.
- Introduction of new goods.
- Unmeasured quality change.

GDP Deflator vs CPI

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GDP Deflator vs CPI

- GDP deflator reflects the prices of all goods produced domestically while CPI focuses on goods bought by consumers.
- Fixed basket with weighting.

Correcting Economic Variables for Effects of Inflation

- Formula:

$$\text{Amount in today's rupees} = \text{Amount in year } T \text{ rupees} \times \frac{\text{Price level today}}{\text{Price level in year } T}$$

- When some dollar amount is automatically corrected for changes in the price level by law or contract, the amount is said to be *indexed* for inflation.

Real and Nominal Interest Rates

$$\text{Real interest rate} = \text{Nominal interest rate} - \text{Inflation rate}$$